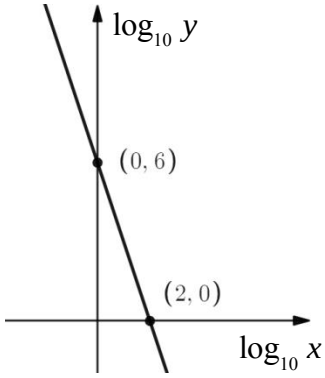
1 (a)	$(5,10)$	B1 B1 **(2)**
(b)	Attempts to solve e.g. $-2(x-5)+10...6x \Rightarrow x...\text{"}\frac{5}{2}\text{"}$ $x < \frac{5}{2}$ o.e.	M1 A1 **(2)**
(c)	$(7, 30)$	B1ft, B1ft **(2)** **(6 marks)**

2	$$g(x)=\frac{2x^2-5x+8}{x-2}$$	
(a)	Sets $2x^2-5x+8=\left(Ax+B\right)(x-2)+C$ with an attempt at one constant	M1
	Attempts all 3 constants. E.g. Sets $x=2\Rightarrow C=...,-2B+C=8\Rightarrow B=...$ and compares x^2 terms gives $A=...$	dM1
	$$2x-1+\frac{6}{x-2}$$	A1
		(3)
(b)	$$\int 2x-1+\frac{6}{x-2}\,dx=x^2-x+6\ln\left(x-2\right)$$	M1 A1ft
	$$\int_4^8 2x-1+\frac{6}{x-2}\,dx=\left[x^2-x+6\ln\left(x-2\right)\right]_4^8=56+6\ln 6-12-6\ln 2$$	
	$$=44+6\ln 3$$	dM1 A1
		(4)
		(7 marks)

3 (i)	$y = \dfrac{10^6}{x^3} \Rightarrow \log_{10} y = \log_{10} 10^6 - \log_{10} x^3 \Rightarrow \log_{10} y = 6 - 3\log_{10} x$  	B1 B1 B1 **(3)**
(ii)	States $\log_3 N = 2t + 4$ **or** $\log_3 N = \log_3 a + t\log_3 b$	B1
	$N = 3^{2t+4} = 3^{2t} \times 3^4$ **or** $\log_3 a = 4 \Rightarrow a = ...$ **or** $\log_3 b = 2 \Rightarrow b = ...$	M1
	$N = 81 \times 9^t$	A1cso **(3)** **(6 marks)**

	$f(x) = 8\sin x\cos x + 4\cos^2 x - 3$	
4(a)	States or uses $\sin 2x = 2\sin x\cos x$ or $\cos 2x = \pm 2\cos^2 x \pm 1$	M1
	Uses $\sin 2x = 2\sin x\cos x$ and $\cos 2x = \pm 2\cos^2 x \pm 1$ in f (x)	dM1
	$\left(f(x) =\right) 8\sin x\cos x + 4\cos^2 x - 3 = 4\sin 2x + 2\cos 2x - 1$	A1
		(3)
(b)	$R^2 = a^2 + b^2 \Rightarrow R = \sqrt{20}$ or $2\sqrt{5}$	B1ft
	$\tan\alpha = \frac{b}{a} \Rightarrow \alpha = ...$ $\left(= \text{"awrt } 0.464\text{"}\right)$	M1
	$\left(f(x) =\right)\ 2\sqrt{5}\sin\left(2x + 0.464\right) - 1$	A1
		(3)
(c)	(i) Maximum value = $\text{"}2\sqrt{5} - 1\text{"}$	B1 ft
	(ii) Solves $2x + \alpha = \frac{5\pi}{2} \Rightarrow x = ...$	M1
	$\left(x =\right)$ awrt 3.69 (or $\left(x =\right)$ awrt 3.70)	A1
		(3)
		(9 marks)

5 (a)	$f^{-1}(22) \Rightarrow 2+5\ln x = 22 \Rightarrow \ln x = 4 \Rightarrow (x=)\ e^4$	M1 A1 **(2)**
(b)	$g(x) = \dfrac{6x-2}{2x+1} \Rightarrow g'(x) = \dfrac{6(2x+1)-2(6x-2)}{(2x+1)^2}$	M1 A1
	States $(g'(x)=)\ \dfrac{10}{(2x+1)^2} > 0$ hence increasing *	A1* **(3)**
(c)	$y = \dfrac{6x-2}{2x+1} \Rightarrow 2xy + y = 6x - 2 \Rightarrow 2xy - 6x = -y - 2$	M1
	$\Rightarrow x = \dfrac{-y-2}{2y-6}$ So $g^{-1}(x) = \dfrac{-x-2}{2x-6}$ o.e.	A1
	Domain $0 < x < 3$	B1 **(3)**
(d)	Range fg is $fg < 2 + 5\ln 3$	M1, A1 **(2)**
		(10 marks)

6 (a)	$y=(4x-7)^{\frac{1}{2}} \quad \Rightarrow\left(\frac{\mathrm{d}y}{\mathrm{d}x}=\right) 2(4x-7)^{-\frac{1}{2}}$	M1 A1
	At $(8,5)$ gradient of tangent is $2(4\times 8-7)^{-\frac{1}{2}}\left(=\frac{2}{5}\right)$	dM1
	Equation for l is $y-5=-\frac{5}{2}(x-8)$	ddM1
	$2y-10=-5x+40 \Rightarrow 5x+2y-50=0$ *	A1*
		(5)
(b)	$\int (4x-7)^{\frac{1}{2}}\,\mathrm{d}x=\left[\frac{(4x-7)^{\frac{3}{2}}}{6}\right]$	M1, A1
	Complete area $=\int_{\frac{7}{4}}^{8}(4x-7)^{\frac{1}{2}}\,\mathrm{d}x=\left[\frac{(4x-7)^{\frac{3}{2}}}{6}\right]_{\frac{7}{4}}^{8}+5$	dM1
	$=\frac{155}{6}$	A1
		(4)
		(9 marks)

7 (a)	$\sqrt{2}\sin(x+45°)=\cos(x-60°)$	
	$\sqrt{2}(\sin x\cos 45°+\cos x\sin 45°)=\cos x\cos 60°+\sin x\sin 60°$	M1 A1
	$\sin x+\cos x=\frac{1}{2}\cos x+\frac{\sqrt{3}}{2}\sin x$	
	$\cos x=(\sqrt{3}-2)\sin x$	M1
	$\tan x\left(=\frac{1}{\sqrt{3}-2}=\frac{\sqrt{3}+2}{-1}\right)=-2-\sqrt{3}$ *	A1*
		(4)
(b)	States or uses $x+45°=2\theta$ o.e.	B1
	Proceeds from e.g. $\tan(2\theta-45°)=-2-\sqrt{3}\Rightarrow 2\theta-45°=105°,285°$	M1
	Correct order of operations to find one angle	dM1
	$\theta=75°,165°$	A1
		(4)
		(8 marks)

8	$h = 1.5x - 0.5x\,\mathrm{e}^{0.02x}$	
(a)	$0 = 1.5d - 0.5d\,\mathrm{e}^{0.02d} \Rightarrow \mathrm{e}^{0.02d} = 3$	M1
	$\Rightarrow 0.02d = \ln 3 \Rightarrow d = 54.93$	dM1 A1
		(3)
(b)	$\left(\frac{\mathrm{d}h}{\mathrm{d}x} =\right) 1.5 - \left(0.5\,\mathrm{e}^{0.02x} + 0.5x \times 0.02\,\mathrm{e}^{0.02x}\right)$	M1A1
	Sets $0 = 1.5 - 0.5\,\mathrm{e}^{0.02x} - 0.5x \times 0.02\,\mathrm{e}^{0.02x} \Rightarrow \mathrm{e}^{0.02x}\left(0.5 + 0.5x \times 0.02\right) = 1.5$	
	$\Rightarrow \mathrm{e}^{0.02x} = \frac{1.5}{\left(0.5 + 0.5x \times 0.02\right)}$	dM1
	$\Rightarrow \mathrm{e}^{0.02x} = \frac{1.5}{\left(0.5 + 0.5x \times 0.02\right)} = \frac{150}{50 + x} \Rightarrow x = 50\ln\left(\frac{150}{50 + x}\right)$ *	A1*
		(4)
(c)	(i) awrt 31.43	M1 A1
	(ii) awrt 30.88 metres **(including units)**	A1
		(3)
		(10 marks)

9 (a)	$(k=)\ 4\sin^2\left(\frac{\pi}{3}\right)-1=2$ *	B1*
		(1)
(b)	(i) $x=4\sin^2 y-1\Rightarrow \quad \frac{\mathrm{d}x}{\mathrm{d}y}=8\sin y\cos y$ o.e.	M1 A1
	(ii) Attempts either $\sin^2 y=\frac{\pm x\pm 1}{4}$ or $\cos^2 y=\pm 1\pm\frac{x+1}{4}$ (both for dM1)	M1 dM1
	$\frac{\mathrm{d}y}{\mathrm{d}x}=\frac{1}{8\sin y\cos y}=\frac{1}{8\times\sqrt{\frac{x+1}{4}}\times\sqrt{\pm 1\pm\frac{x+1}{4}}}=\frac{1}{2\sqrt{x+1}\sqrt{3-x}}$ *	ddM1 A1*
		(6)
(c)	At $x=2$, gradient of curve $=\frac{1}{2\sqrt{3}}\Rightarrow$ gradient of normal is $-2\sqrt{3}$	M1
	Point N (base length of triangle) is solution of $x-\frac{\pi}{3}=-2\sqrt{3}\left(x-2\right)\Rightarrow x=2+\frac{\pi}{6\sqrt{3}}$ $(=2.3...)$	dM1
	Area $=\frac{1}{2}\times\left(2+\frac{\pi}{6\sqrt{3}}\right)\times\frac{\pi}{3}=\frac{\pi}{3}+\frac{\pi^2}{36\sqrt{3}}$	A1
		(3)
		(10 marks)